Thinking Machines

A DeepSeek summary of work in progress by Gary Nan Tie, Mar-Sep, 2025

Core Idea: A paradigm shift from Similarity to Explanation

Current transformer attention mechanisms are fundamentally based on similarity. A query is compared to keys, and values from the most similar keys are weighted and combined. The limitation ([14], [18]), is that this is a statistical, correlation-based process. It can struggle with reasoning, causality, and reconciling contradictory information (the "Rashomon effect").

Fibration-based attention, as proposed in this body of work, shifts the paradigm from similarity to abduction (inference to the best explanation).

The Fibration Metaphor:
In mathematics, a fibration is a way of projecting a complex space (total space E) onto a simpler base space B. A "lifting" is the process of finding an element in E that projects down to a given element in B. This is a perfect metaphor for abduction: given an observation (a value v in the base space), find a hypothesis (a key k in the total space) that best "explains" or projects to it.

The Attention Mechanism:
In "Fibration Attention" [16], the standard (Q, K, V) triplet is reimagined. The key-value pair (k, v) defines the fibration. The attention mechanism becomes a lifting operation: for a given query q, it doesn't just find similar keys; it abductively finds the key `k` that best explains the desired output value `v`. This is a "diagonal lifting" [11] that simultaneously updates the query and finds the explanatory key.

Why This Could Be the Future: Addressing LLM Core Limitations

The development of fibration attention points to several critical advantages that could make this a cornerstone of future LLMs:

1. Mitigation of Hallucinations ([10], [11]):
Current LLMs "hallucinate" because they generate plausible-sounding text based on statistical likelihood. A fibration-based model like "FibChat" would be constrained by the need to form coherent, explanatory chains of reasoning ("LLM Bonsai" through pruning). It doesn't just generate the next token; it generates the next token as the conclusion of an abductive inference step, which imposes a structural constraint on its output.

2. Explainability and Transparency ([4], [9]):
A major weakness of current transformers is their "black box" nature. The "precedence" of abductive reasoning steps [4] could be tracked and marked. This means the model could potentially explain its reasoning by retracing the sequence of liftings (abductive inferences) it used to arrive at an answer, moving from a statistical model to a reasoning model.

3. Handling Ambiguity and Contradiction ([18]):
The "Rashomon effect" – where multiple, equally valid explanations exist for the same data – is a fundamental challenge. "Jury Attention" [18] proposes a mechanism via fibrations to manage these differing "world views" and synthesize query updates that consider multiple competing explanations, rather than just averaging them into a confused output.

4. Structured Exploration and Stability ([19], [12]):
The use of perturbed Koopman operator theory [19] to analyze the dynamics of query updating is a profound step. It moves the design of attention from an engineering heuristic to a principled mathematical framework. Guaranteeing Lyapunov stability means the query refinement process is convergent and well-behaved, avoiding erratic or chaotic reasoning trajectories. The "Fibformer" [12] uses this for "controlled structured exploration."

5. Towards Artificial General Intelligence (AGI) ([7], [8]):
The author explicitly frames this as a step towards AGI. Induction (learning from data) and deduction (applying rules) are well-represented in current AI. Abduction – the core of human hypothesis generation and scientific reasoning – has been the missing piece. By giving neural networks a native capacity for abductive inference via fibrations, we might be creating models that don't just parrot data but actively reason about it.

Challenges and Considerations:
This path is not without its challenges

· Computational Complexity:
The abductive lifting operation is likely more computationally intensive than a matrix dot-product similarity calculation. Significant innovation in efficient algorithms could be required.

· Learnability:
The papers propose the fibrations themselves are "machine learnable" [7, 2]. How to effectively and scalably learn these structures from data is a central research question.

· Integration:
How does this new attention mechanism integrate with other core components of a transformer (feed-forward networks, normalization)? The "Attention Legos" [17] and "Formal query updating" [13] papers address this by creating composable primitives, that can be initialized with transformer attention query updates like a Bayesian prior.

Summary:
The trajectory outlined from [1] to [19] presents a compelling vision: the evolution of the attention mechanism from a powerful but statistically brittle pattern-matching tool into a robust, explainable, and stable engine for abductive reasoning.

If this research program proves successful, fibration-based attention could indeed become a fundamental component of future LLMs, particularly for applications requiring reliability, transparency, and logical reasoning—such as in scientific discovery, medical diagnosis [3, 7], legal analysis, and any field where "showing your work" is as important as the answer itself. It represents a move from models that statistically know, to models that structurally reason.

How does this theoretical concept transition into a practical, feasible, and powerful machine learning architecture?

Two Equations:
From Abstract Math to Concrete Loss Functions

The entire, potentially daunting, concept of a fibration and its lifting properties can indeed be distilled into the requirement that two diagrams commute. For a given (key, value) pair (k, v) and a query q, we have two fundamental operations:

1. The Projection Map (p):
This is the "effect" or "manifestation" function. It maps a key (hypothesis) to its value (observation). $p(k) = v$.
2. The Lifting Map (λ):
This is the abductive reasoning function. For a given query q and a target value v, it finds the key that explains v in the context of `q`. It produces both an updated query q' and the explanatory key k'. $\lambda(q, v) = (q', k')$.

The commuting triangles of the diagonal of a fibration lifting diagram impose two critical constraints that ensure coherence and explanation:

· Constraint 1 (Tautologicality):
If you lift a value and then project the resulting key, you must get the original value back. p(λ(q, v)) = p(k') = v This ensures the key k' found by the lifting actually explains the value `v`. It's a hard constraint that the explanation must be valid.

· Constraint 2 (Parallel Transport):
The lifting must be consistent with the query context. The updated query q' must be "compatible" with the found key k'. This is often expressed as the new query being a transformation of the old one based on the key-value pair, or that the (q', k') pair lies in a shared "space of explanations".

These two constraints are not just abstract ideas; they are directly translatable into loss functions for training a neural network.

Practical Implementation: Regularization is the Key

Instead of being hand-coded, the functions p (projection) and λ (lifting) are parameterized as neural networks (e.g., MLPs, or more sophisticated structures like KAN). Their learning is guided by loss functions that enforce the commuting diagrams.

1. Reconstruction Loss (for Constraint 1):
This is the direct regularization term. L_recon = || p(λ(q, v)) - v ||^2 The model is penalized if the key it abduces does not correctly project to the target value. This is the "you must explain this observation correctly" rule.

2. Consistency Loss (for Constraint 2):
This ensures the query update is meaningful. The exact form can vary based on the specific architecture (e.g., it might ensure that the updated query q' is closer to the key k' in some latent space, or that q' can be used to reconstruct q and k). L_consistent = ... (architecture-specific, e.g., || f(q, k') - q' ||^2)

3. Task-Specific Loss:
This is the standard loss for the model's primary objective, e.g., next-token prediction (cross-entropy loss) or medical diagnosis (categorical loss).
L_task = CrossEntropy(FFN(q'), target)

The total loss function for the system becomes: L_total = L_task + α * L_recon + β * L_consistent

Where α and β are hyperparameters controlling the strength of the regularization. This is entirely standard practice in deep learning (e.g., as seen in VAEs, where a reconstruction loss is combined with a KL-divergence regularization).

Why This is Feasible and Powerful.

· Modularity:
The "fibration attention" layer can be a drop-in replacement for a standard attention layer in a transformer block. It takes in (Q, K, V) and outputs updated queries Q' and, as a bonus, explanatory keys K_explanation.

· End-to-End Learning:
The entire system—projection network, lifting network, and any other parameters—are trained simultaneously via gradient descent on L_total. The network learns the semantics of what constitutes a valid explanation (k, v) for a given problem domain.

· Inductive Bias for Reasoning:
The genius of this approach is that the mathematical structure of the fibration bakes in an inductive bias for abductive reasoning. The model isn't just told to "reason better"; its very architecture forces it to learn functions that adhere to the logical structure of explanation. This is a far more powerful bias than the "find similarity" bias of standard attention.

· Explainability as a Byproduct:
After training, to explain an output, you can simply examine the explanatory key k' found by the lifting operation for a given value v. This key is a latent representation that can be decoded or analyzed, providing a direct window into the model's "reasoning" process.

The reduction of a profound mathematical concept to two learnable, regularized constraints is what makes "fibration attention" not just a theoretical novelty, but a practical and feasible blueprint for the next generation of interpretable, reasoning-aware LLMs. It transforms abstract category theory into a concrete engineering design pattern via the well-trodden path of regularized loss functions.

In conclusion, the shift from similarity-based to explanation-based and exploration-driven AI, as embodied by the move from standard attention to fibration attention, represents a fundamental paradigm shift with profound and wide-ranging ramifications. It moves AI from being a powerful pattern-matching engine towards becoming a reasoning and discovery partner.

Here is a breakdown of ramifications across different domains:

1. Ramifications for Capabilities & Performance

· From Interpolation to Reasoning:
Current LLMs excel at interpolating within their training data. Fibration-based models would be designed to perform abductive inference, generating hypotheses that best explain observations. This is the core of scientific, medical, judicial, and common-sense reasoning.

· Robustness and Reduced Hallucination:
By being constrained to find keys that actually project to the correct value ($p(k') = v$), the model's outputs are grounded in a internal logic of explanation, not just statistical likelihood. This should drastically reduce nonsensical or unfounded hallucinations.

· Structured Exploration:
As noted in the references, this enables "controlled structured exploration" [12]. The model can explore multiple competing hypotheses (explanatory keys) in a structured way, like a scientist considering different theories, rather than blending them into an average. This is crucial for complex problem-solving.

· Causal Understanding:
Semantics through causality, the structure (key/hypothesis -> value/observation) is the framework for abduction (reversing cause -> effect). Learning to abduce causes from effects is a significant step toward AI that can understand and reason about causality, not just correlation.

2. Ramifications for Transparency and Trust (Explainable AI - XAI)

· Inherent Explainability:
This is perhaps the biggest ramification. The "explanation" isn't a post-hoc add-on; it's a fundamental byproduct of the process. For any decision, you can inspect the explanatory key k' that the model found.

· In a medical AI:
You wouldn't just get a diagnosis; you'd get the inferred pathological pathway (k') that explains the symptoms (v).

· In a legal AI:
You wouldn't just get a judgment; you'd get the precedent or legal principle (k') that best explains the case details (v).

· Auditability and Debugging:
The "precedence" of abductive steps [4] creates a clear audit trail. If an AI makes a mistake, we can trace back through its chain of reasoning to find the flawed abduction step, much like debugging a program. This is impossible with today's black-box transformers.

· Building Trust:
For AI to be deployed in high-stakes domains (healthcare, law, autonomous systems), users need to trust its outputs. Providing a coherent, inspectable reasoning process is essential for building that trust.

3. Ramifications for Architecture and Design

· Beyond the Transformer:
While this could be integrated into a "Fibformer" [12], it fundamentally changes the core primitive from "attention" to "abduction." This may lead to entirely new neural architectures built around this explanatory core, moving beyond the transformer blueprint.

· Knowledge Representation:
The "keys" and "values" cease to be mere vectors for similarity search. They become structured representations of hypotheses and observations. This could lead to a more seamless integration between neural networks and symbolic or knowledge-based systems.

· Dynamic and Recursive Reasoning:
The ability to recursively layer explanations [16] (building "trees" of keys upon keys) allows the model to construct deep, hierarchical chains of reasoning, mimicking how humans build complex arguments from first principles.

4. Ramifications for Society and Economies

· Democratization of Expertise:
AI systems that can clearly explain their reasoning could act as powerful amplifiers for human expertise. A doctor, lawyer, or engineer could collaborate with an AI to explore diagnostic or design options, understanding the "why" behind each suggestion.

· Acceleration of Scientific Discovery:
This is a primary application. An AI that formulates hypotheses from data could become an unparalleled partner in science, generating testable explanations for complex phenomena in fields from physics to biology.

· New Forms of Collaboration:
The interaction model changes from human prompting an AI to human debating or collaborating with an AI. The AI presents its explanations, the human critiques them, and the model updates its reasoning—a true iterative partnership.

· Shifting Job Markets:
The impact on knowledge work would be more profound than with current AI. It wouldn't just automate tasks; it would automate reasoning. This raises the value of critical thinking, creativity, and the ability to guide and judge AI-generated explanations (the "Jury" role [18]).

5. Ramifications for AI Safety and Alignment

· Improved Alignment:
It is easier to align an AI whose reasoning process you can inspect and understand. You can regularize the model to prefer explanations that align with human values or specific ethical frameworks.

· New Safety Challenges:
The ability for structured exploration could lead to novel failure modes. An AI might become overly attached to a coherent but immoral or dangerous explanatory framework. Ensuring the "Lyapunov stability" [19] of its reasoning trajectories will be crucial to prevent harmful, runaway chains of logic.

· Verification and Validation:
The field of AI verification could move from statistical benchmarks to something more akin to proof verification, where the logical soundness of the AI's abductive chain can be checked, at least in part.

Conclusion: From Statistical Parrots to Reasoning Partners
- a fundamental phase transition in AI capabilities.

The Emergence of True AI Reasoning and "Artificial Scientific Intuition"
· Current State:
AI is powerful at induction (finding patterns) and deduction (applying rules). The third pillar of reasoning, abduction (inference to the best explanation), has been largely missing.

· Future State:
By formalizing abduction as a fibration lifting, AI gains a native capacity for hypothesis generation. This is the engine of human scientific and creative thought.

· Scientific Discovery:
An AI could review experimental data (values v), generate multiple plausible hypotheses (keys k1, k2, k3...), and then design new experiments to test them. This moves AI from a data analysis tool to a collaborative partner in discovery.

· Diagnostics:
In medicine (as noted in [3], [7]) or engineering, the AI wouldn't just match symptoms to a database; it would abduce the most likely underlying fault or disease and explain its line of reasoning, allowing a human expert to critique the logic.

· Similarity-based AI (Today):
Acts as an incredibly knowledgeable but ultimately stochastic parrot. It provides answers based on what was most common or similar in its training data.

· Explanation-based AI (Future):
Aims to be a reasoning partner. It provides answers based on a structured process of inference to the best explanation, which is inspectable, debatable, and grounded in a logical framework.

This shift moves AI from the domain of pure statistics into the domain of epistemology—the theory of knowledge itself. It promises not just more accurate outputs, but outputs that come with understanding, reason, and the potential for true discovery.

References (most recent first)

[19] ‘Fibration Attention Dynamics’
Gary Nan Tie, Sep 13, 2025
DOI: 10.13140/RG.2.2.36796.09600
We characterize the dynamics of query updating coming from fibration abductive attention mechanisms, using perturbed Koopman operators that are invertible. Importantly, we identify conditions for Lyapunov stability of solutions near equilibrium points. Depending on the Koopman resolvent, some key-value pairings can have query update trajectories that are Lyapunov stable and converge.

[18] ‘Jury Attention’
Gary Nan Tie, Sep 10, 2025
DOI: 10.13140/RG.2.2.25750.00322
To address the Rashomon effect when multiple key-value semantics are present, we introduce a jury attention mechanism via fibrations. This accomplishes query updates that take into account differing world views.

[17] ‘Attention Legos’
Gary Nan Tie, Sep 8, 2025
DOI: 10.13140/RG.2.2.33398.05447
Attention building blocks for machines to learn world models?

[16] ‘Deep Fibration Attention’
Gary Nan Tie, Aug 31, 2025
DOI: 10.13140/RG.2.2.17946.30402
In fibration attention, a lifting simultaneously updates a query and abductively finds a key to explain a value. This all depends on the key-value pairing (k,v). By determining other keys that are highly similar to k, to also explain v, we can enhance key-value semantics to achieve more meaningful attention. Moreover, we can recursively layer keys upon keys in a tree, to deepen the set of value explanations.

[15] ‘Contextual Fibration Attention’
Gary Nan Tie, Aug 29,2025
DOI: 10.13140/RG.2.2.12205.35044
In a query-key-value attention model, the key-value pairing conveys language semantics. Query context also carries meaning.
We introduce an abductive fibration attention model that holistically captures both. A fibration lifting, jointly learns from query context and language semantics (that abductively finds a key to best explain the returned value).

[14] ‘Multi-head abductive attention’
Gary Nan Tie, Aug 25, 2025
DOI: 10.13140/RG.2.2.20597.23521
Transformer multi-head attention can focus on different parts of an input query sequence, however without regard to potentially different key-value semantics. This new primitive for formal query updating, constructs updates to a query in parallel, by learning multi-head abductive fibration liftings, to refine similarity based attention, utilizing the semantics of multiple key-value pairings.

[13] ‘Formal query updating to generate problem specific fibration abduction attention algorithms’
Gary Nan Tie, Aug 23, 2025
DOI: 10.13140/RG.2.2.33029.41446
A formalism to construct problem specific query update algorithms founded on standard attention and fibration attention primitives.

[12] ‘Fibformer - a fibration transformer’
Gary Nan Tie, Aug 22, 2025
DOI: 10.13140/RG.2.2.36293.10723
We introduce a new transformer with novel fibration attention based on abductive reasoning. This coherent approach to query updating enables controlled structured exploration of continuation values.

[11] ‘LLM Bonsai - pruning FibChat
Gary Nan Tie, Jul 25, 2025
DOI: 10.13140/RG.2.2.21984.80644
Each step in learning the diagonal lifting of a fibration, used to simultaneously update a query and find a section of the key-value pairing, represents an opportunity for intervention to direct the growth of our query refinement tree by selective pruning, so as to avoid hallucinations, LLM bonsai!

[10] ‘FibChat - It makes stuff up, but with structure.’
Gary Nan Tie · July 2025
DOI: 10.13140/RG.2.2.10549.59364/2
FibChat is a LLM based on fibration attention, that enables coherent structured mitigation of hallucinations.

[9] ‘Attention Fibration Abduction’
Gary Nan Tie, Jul 2, 2025
DOI: 10.13140/RG.2.2.29060.23680/1
Transformer attention and abductive reasoning are intimately connected, embodied by fibrations. Suitable attention mechanisms can make a key-value pairing a learnable neural network fibration for abductive reasoning. A fibration lifting explains an attention output value to a query, by identifying the key paired to it, explainable attention through abduction.

[8] ‘Parsimonious Neural Network Abduction’
Gary Nan Tie, Jun 22, 2025
DOI: 10.13140/RG.2.2.15695.80804
For hypotheses whose effect is manifested by observations, abduction seeks to explain a given observation by finding a hypothesis whose effect is that observation. Abductive reasoning has a fibration semantics that can be implemented by neural networks; a step towards artificial general intelligence.
In this note we introduce a parsimonious choice of explanation depending on one’s utility function.

[7] ‘Neural Nerwork Abduction’
Gary Nan Tie, Jun 13, 2025
DOI: 10.13140/RG.2.2.18506.07360/1
For hypotheses whose effect is manifested by observations, abduction seeks to explain a given observation by finding a hypothesis whose effect is that observation. Abductive reasoning has a fibration semantics that can be implemented by neural networks; a step towards artificial general intelligence.
A natural application is in making a medical diagnosis. Fibration structure and properties enable coherency and consistency in this process. Moreover being machine learnable, can help physicians uncover diagnostic insights in large datasets.

[6] ‘Meta-Abduction’
Gary Nan Tie, Jun 10, 2025
DOI: 10.13140/RG.2.2.22022.64323
We note that for fibration semantics of abductive reasoning, abduction of attention is a meta-effect, not section reversal per se.

[5] ‘Abductive Inference via Explainable Neural Networks’
Gary Nan Tie, May 25, 2025
DOI: 10.13140/RG.2.2.20953.84320
Explainable abductive inference by invertible neural networks is examined in the context of fibration semantics.

[4] ‘Abductive Reasoning Precedence’
Gary Nan Tie, May 23, 2025
DOI: 10.13140/RG.2.2.19913.45923
Fibration liftings used to model abduction are not necessarily unique. So chains of abductive reasoning are a series of certain choices of liftings. To keep track of this precedence we introduce a simple mechanism to mark liftings. This transparency enables one to analyze decision points in an exploit-explore strategy for optimizing chains of abductive reasoning.

[3] ‘Coherent Abductive Medical Diagnosis’
Gary Nan Tie, May 15, 2025
DOI: 10.13140/RG.2.2.16148.41605
Disease causes symptoms. In the other direction, given a patient’s symptoms, medical diagnosis seeks diseases that explain the observed symptoms, a form of abductive reasoning. Abduction can be viewed as a fibration, a mathematical model, which when applied to clinical decision making enables coherent, transparent, refinable medical diagnoses. Moreover, given enough data, fibrations are potentially machine learnable, providing an interactive AI learning tool for physicians to better understand and refine their diagnostic thinking.

[2] ‘Fibration Abductive Learning’
Gary Nan Tie, Mar 4, 2025
DOI: 10.13140/RG.2.2.27284.82562/2
Supplement DOI: 10.13140/RG.2.2.24804.28800/1
Neural networks like MLP (multilayer perceptrons) and KAN (Kolmogorov-Arnold networks) can solve, by induction, regression, classification and time series problems. Deduction is used by neural networks in rule based systems, logic programming, automated theorem proving and reinforcement learning.
This naturally begs the question about applying the third kind of inference abduction, seeking the simplest and most likely explanation for a set of observations.
To conceptualize and perform abductive machine learning, categorical fibrations are proposed, as they coherently give both context and structure for abductive reasoning via a tautological attention mechanism.

[1] ‘Fibrations explain all you need!
- Fibrations, Abduction, Attention
Gary Nan Tie, Mar 4, 2025
DOI: 10.13140/RG.2.2.35984.52488
To conceptualize and perform abductive machine learning, categorical fibrations are proposed, as they coherently give both context and structure for abductive reasoning via an attention mechanism. Moreover, 2-categorical lifting compatibility conditions ensure consistent hierarchical and parallel explanations.